

Python DICTIONARY HOMEWORK

Q1. Loop Through Dictionary and Remove Items

Given:

```
students = {"Ali": 45, "Sara": 90, "Rohan": 29, "Meera": 50}
```

Remove all students scoring **less than 40**
without modifying the dictionary while looping over it.

Q2. Create a Dictionary from Two Lists

Input:

```
keys = ["name", "age", "city"]  
values = ["Ali", 20, "Delhi"]
```

Create a dictionary using these lists.
But if lists have **different lengths**, print: "Cannot create dictionary".

Q3. Reverse Dictionary

Given the dictionary:

```
country_capital = {"India": "Delhi", "France": "Paris", "Japan":  
"Tokyo"}
```

Write a Python program to **create a new dictionary where keys become values and values become keys**.

Q4. Merge Two Dictionaries Without Using `update()`

Given:

```
d1 = {"a": 1, "b": 2}  
d2 = {"b": 3, "c": 4}
```

Write a program to **manually merge** the dictionaries so that if a key exists in both, the value from **d2 overrides** the value in **d1**.

Q5. Dictionary of Squares (Even → Cube)

Write a program that takes a number **n** and creates a dictionary where:

- odd numbers store **their square**
- even numbers store **their cube**

Example for **n = 4**:

```
{1: 1, 2: 8, 3: 9, 4: 64}
```

Q6. Remove Students Scoring Less Than 40

Given:

```
students = {"Ali": 45, "Sara": 90, "Rohan": 29, "Meera": 50}
```

Write a program to **remove all students who scored below 40**.

Q7. Create Dictionary from Two Lists

Given two lists:

```
keys = ["name", "age", "city"]  
values = ["Ali", 20, "Delhi"]
```

Write a program to create a dictionary by **matching each key with the corresponding value**.

If lengths do not match → print an error.